



Inequalities between s-numbers

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Abstract

Singular numbers of linear operators between Hilbert spaces were generalized to Banach spaces by s-numbers (in the sense of Pietsch). This allows for different choices, including approximation, Gelfand, Kolmogorov and Bernstein numbers. Here, we present an elementary proof of a bound between the smallest and the largest s-number.

Keywords s-Numbers · Hilbert numbers · Pietsch

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We start with introducing the terminology and a presentation of the results. In Sect. 2 we will discuss them and some history. Proofs are given in Sect. 3.

1 s-Numbers

In what follows, let X , Y , Z and W be real or complex Banach spaces. The (closed) unit ball of X is denoted by B_X , the dual space of X by X' , and the identity map on X is denoted by I_X . For a closed subspace $M \subset X$, we write J_M^X for the embedding $J_M^X: M \rightarrow X$ with $J_M^X(x) = x$, and Q_M^X for the canonical map $Q_M^X: X \rightarrow X/M$ with $Q_M^X(x) = x + M$ onto the quotient space $X/M := \{x + M: x \in X\}$ with norm $\|x + M\|_{X/M} := \inf_{m \in M} \|x + m\|_X$. The dimension of a subspace $M \subset X$ is denoted by $\dim(M)$, and by $\text{codim}(M) := \dim(X/M)$ we denote its codimension.

The class of all bounded linear operators between Banach spaces is denoted by $\mathcal{L}$, and by $\mathcal{L}(X, Y)$ we denote those operators from X to Y , equipped with the operator norm. The rank of an operator $S \in \mathcal{L}(X, Y)$ is defined by $\text{rank}(S) := \dim(S(X))$.

In memory of Albrecht Pietsch

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A map $S \rightarrow (s_n(S))_{n \in \mathbb{N}}$ assigning to every operator $S \in \mathcal{L}$ a non-negative scalar sequence $(s_n(S))_{n \in \mathbb{N}}$ is called an *s-number sequence* if, for all $n \in \mathbb{N}$, the following conditions are satisfied.

- (S1) $\|S\| = s_1(S) \geq s_2(S) \geq \dots \geq 0$ for all $S \in \mathcal{L}$,
- (S2) $s_n(S + T) \leq s_n(S) + \|T\|$ for all $S, T \in \mathcal{L}(X, Y)$,
- (S3) $s_n(BSA) \leq \|B\| s_n(S) \|A\|$ where $W \xrightarrow{A} X \xrightarrow{S} Y \xrightarrow{B} Z$,
- (S4) $s_n(I_{\ell_2^n}) = 1$,
- (S5) $s_n(S) = 0$ whenever $\text{rank}(S) < n$.

We call $s_n(S)$ the n^{th} *s-number* of the operator S . To indicate the underlying Banach spaces, we sometimes write $s_n(S: X \rightarrow Y)$.

There are some especially important examples of s-numbers:

- approximation numbers:

$$a_n(S) := \inf \left\{ \|S - L\| : L \in \mathcal{L}(X, Y), \text{rank}(L) < n \right\}$$

- Bernstein numbers:

$$b_n(S) := \sup \left\{ \inf_{x \in M \setminus \{0\}} \frac{\|Sx\|}{\|x\|} : M \subset X, \dim(M) = n \right\}$$

- Gelfand numbers:

$$c_n(S) := \inf \left\{ \|SJ_M^X\| : M \subset X, \text{codim}(M) < n \right\}$$

- Kolmogorov numbers:

$$d_n(S) := \inf \left\{ \|Q_N^Y S\| : N \subset Y, \dim(N) < n \right\}$$

- Weyl numbers:

$$x_n(S) := \sup \left\{ \frac{a_n(SA)}{\|A\|} : A \in \mathcal{L}(\ell_2, X), A \neq 0 \right\}$$

- Hilbert numbers:

$$h_n(S) := \sup \left\{ \frac{a_n(BSA)}{\|B\| \|A\|} : A \in \mathcal{L}(\ell_2, X), B \in \mathcal{L}(Y, \ell_2), A, B \neq 0 \right\}.$$

We refer to [13, 14] for a detailed treatment of the above, and a few other, s-numbers and their specific properties.

Remark 1 The original definition of s-numbers in [11] used the stronger *norming property* (S4'): $s_n(I_X) = 1$ for all X with $\dim(X) \geq n$. This did not allow for x_n and h_n , and has been weakened in [1, 12] for defining them, leading to the least

restrictive axioms that still imply uniqueness for Hilbert spaces, see Proposition 2. It is sometimes assumed that the s-number sequence is *additive*, i.e., $(S2')$: $s_{m+n-1}(S+T) \leq s_m(S) + s_n(T)$, or *multiplicative*, i.e., $(S3')$: $s_{m+n-1}(ST) \leq s_m(S)s_n(T)$. Both properties hold for a_n, c_n, d_n, x_n , while h_n is only additive, and b_n is neither of the two, see [14, 15].

The following proposition is well-known, see [13, 2.3.4 & 2.6.3 & 2.11.9].

Proposition 2 *For every s-number sequence (s_n) , $S \in \mathcal{L}$ and $n \in \mathbb{N}$, we have*

$$h_n(S) \leq s_n(S) \leq a_n(S).$$

Equalities hold if $S \in \mathcal{L}(H, K)$ for Hilbert spaces H and K .

For convenience, we present a sketch of the proof of the inequalities in Sect. 3. Using only elementary arguments, we prove the following reverse inequality, which is known in a more involved form based on an intermediate comparison with $x_n(S)$, see [13, 2.10.7] or [14, 6.2.3.14], or Remark 6 in [6].

Theorem 3 *For all $S \in \mathcal{L}$ and $n \in \mathbb{N}$, we have*

$$\max\{c_n(S), d_n(S)\} \leq n \left(\prod_{k=1}^n h_k(S) \right)^{1/n}.$$

Since $c_n(I: \ell_1 \rightarrow \ell_\infty) \asymp 1$ and $h_n(I: \ell_1 \rightarrow \ell_\infty) \asymp n^{-1}$, see [14, 6.2.3.14] and [3], this result is best possible up to constants.

We cannot obtain bounds for individual n from Theorem 3, see also Remark 5, but combining it with the inequality $n^\alpha \leq e^\alpha (n!)^{\alpha/n}$ for $\alpha \geq 0$, we obtain a more handy form. Moreover, the known fact that $a_n(S) \leq (1 + \sqrt{n}) c_n(S)$, see [13, 2.10.2], leads to a bound between a_n and h_n , and hence between all s-numbers.

Corollary 4 *For all $S \in \mathcal{L}$, $\alpha > 0$ and $n \in \mathbb{N}$, we have*

$$c_n(S) \leq e^\alpha n^{-\alpha+1} \cdot \sup_{k \leq n} k^\alpha h_k(S).$$

and

$$a_n(S) \leq 2 e^\alpha n^{-\alpha+3/2} \cdot \sup_{k \leq n} k^\alpha h_k(S).$$

2 A bit of history

We provide a brief description of the relevant facts from (the highly recommended) “History of Banach Spaces and Linear Operators” of Pietsch [14], and give some further references.

Singular numbers of operators on Hilbert spaces have become fundamental tools in (applied) mathematics since their introduction in 1907 by Schmidt [18]. For compact $S \in \mathcal{L}(H, K)$ between complex Hilbert spaces H, K , the *singular numbers* are defined by $s_k(S) := \sqrt{\lambda_k(SS^*)}$, where the *eigenvalues* $\lambda_k(T)$ of $T \in \mathcal{L}(X, X)$ are characterized by $Te_k = \lambda_k(T) \cdot e_k$ for some $e_k \in X \setminus \{0\}$, and ordered decreasingly. Applications range from the study of *eigenvalue distributions* of operators, see [5] or [14, 6.4], to the classification of *operator ideals* [14, 6.3], to the singular value decomposition, aka Schmidt representation, with its many applications.

s-numbers are a generalization to linear operators between Banach spaces. However, there is no unique substitute for singular numbers but, depending on the context, different s-numbers may be used to *quantify compactness*, while others may be easier to compute. As Pietsch wrote in [14, 6.2.2.1], “we have a large variety of s-numbers that make our life more interesting.”

Most notably, a_n, b_n, c_n and d_n were already known in the 1960s, sometimes in a related form as “width” of a set, see [4, 20] or [14, 6.2.6], and are by now part of the foundation of approximation theory [17] and information-based complexity [8, 10]. More recent treatments of the subject and extensions can be found, e.g., in [2, 7, 19]. Let us also highlight [6], where we discuss the relation of s-numbers to *minimal approximation errors* in detail, and use variants of Theorem 3 to bound the *maximal gain* of randomized/adaptive algorithms over deterministic/non-adaptive ones.

An axiomatic theory of s-numbers has been developed by Pietsch [11, 13, 14] in the 1970s. This, in particular, allowed for a characterization of the smallest/largest s-number (with certain properties), see also Remark 1.

Inequalities and several relations between s-numbers have already been collected in [11], see also [13, 2.10] and [14, 6.2.3.14]. A particularly interesting bound is $d_n(S) \leq n^2 b_n(S)$, which was proven by Mityagin and Henkin [9] in 1963. They also conjectured that n^2 can be replaced by n , see also [17, p. 24]. This bound with b_n replaced by h_n , and the corresponding conjecture, have been given in [1], and apparently, the bound has not been improved since then. However, in the weaker form as in Theorem 3, this problem has been solved by Pietsch [12] in 1980, see the final remarks there. In particular, it is shown that there is some $C_\alpha > 0$ such that $\sup_{k \geq 1} k^\alpha d_k(S) \leq C_\alpha \sup_{k \geq 1} k^{\alpha+1} h_k(S)$, see also [13, 2.10.7] or [14, 6.2.3.14].

The proof of this result is the blueprint for the proof of Theorem 3. However, those proofs employ an intermediate comparison with the Weyl numbers x_n , which lie somehow between h_n and c_n . Despite interesting consequences, this approach requires the multiplicativity of x_n , the notion of 2-summing norm, and some technical difficulties. The proof presented here is elementary: It only uses definitions and known properties of the determinant.

Bounds for individual n cannot be deduced from this approach, see also Remark 5, and it remains a long-standing open problem if $d_{bn}(S) \leq c n b_n(S)$, or even $a_{bn}(S) \leq c n h_n(S)$, for some $b, c \geq 1$, see [16, Prob. 5] or [13, 2.10.7].

3 The proofs

Proof of Proposition 2 We refer to [13, 2.11.9] for the proof that $s_n(S) = a_n(S)$ for any s-number sequence (s_n) , and any $S \in \mathcal{L}(H, K)$ for Hilbert spaces H and K . Just note that, for compact S , this follows quite directly from the singular value decomposition. From this and (S3), we obtain

$$h_n(S) = \sup \left\{ \frac{s_n(BSA)}{\|B\| \|A\|} : A \in \mathcal{L}(\ell_2, X), B \in \mathcal{L}(Y, \ell_2), A, B \neq 0 \right\} \leq s_n(S)$$

for any (s_n) . In addition, by (S2) and (S5), we obtain for any L with $\text{rank}(L) < n$ that $s_n(S) \leq s_n(L) + \|S - L\| = \|S - L\|$. By taking the infimum over all such L , we see that $s_n(S) \leq a_n(S)$. $\square$

Proof of Theorem 3 We first present the proof from [13, 2.10.3] of the following statement: For fixed $\varepsilon > 0$, we can find $x_1, \dots, x_n \in B_X$ and $b_1, \dots, b_n \in B_{Y'}$ such that $\langle Sx_k, b_j \rangle = 0$ for $j < k$ and $(1 + \varepsilon)|\langle Sx_k, b_k \rangle| > c_k(S)$ for $k = 1, \dots, n$.

For this, we inductively assume that x_k, b_k for $k < n$ are already found, and define

$$M_n := \left\{ x \in X : \langle Sx, b_k \rangle = 0 \text{ for } k < n \right\}.$$

Since $\text{codim } M_n < n$, we can choose $x_n \in M_n \cap B_X$ with

$$(1 + \varepsilon)\|Sx_n\| \geq \|SJ_{M_n}^X\| \geq c_n(S).$$

By the Hahn-Banach theorem, we choose $b_n \in B_{Y'}$ with $\langle Sx_n, b_n \rangle = \|Sx_n\| \geq \frac{c_k(S)}{1+\varepsilon}$.

We now define the operators

$$A(\xi) := \sum_{i=1}^n \xi_i x_i \in X, \quad \xi = (\xi_i) \in \ell_2^n,$$

and

$$B(y) := \left(\langle y, b_i \rangle \right)_{i=1}^n \in \ell_2^n, \quad y \in Y,$$

which satisfy $\|A\|, \|B\| \leq \sqrt{n}$, and observe that $S_n := BSA: \ell_2^n \rightarrow \ell_2^n$ is generated by the triangular matrix $(\langle Sx_i, b_j \rangle)_{i,j=1}^n$ with determinant $\det(S_n) \geq \prod_{k=1}^n \frac{c_k(S)}{1+\varepsilon}$.

To obtain a bound with s-numbers, note that they all coincide for S_n , esp. with $a_k(S_n)$, and are equal to the singular numbers $s_k(S_n)$, i.e., the roots of the eigenvalues of $S_n S_n^*$, see [14, 6.2.1.2]. As the determinant is multiplicative and equals the product of the eigenvalues, we see that $\det(S_n) = \sqrt{\det(S_n S_n^*)} = \prod_{k=1}^n a_k(S_n)$. From the definition of h_n , we obtain $a_k(S_n) \leq \|A\| \|B\| h_k(S) \leq n \cdot h_k(S)$, and hence

$$(1 + \varepsilon)^{-n} \prod_{k=1}^n c_k(S) \leq \det(S_n) = \prod_{k=1}^n a_k(S_n) \leq n^n \prod_{k=1}^n h_k(S).$$

With $\varepsilon \rightarrow 0$ and $c_n(S) \leq (\prod_{k=1}^n c_k(S))^{1/n}$ we obtain the result for $c_n(S)$.

The proof for $d_n(S)$ could be done via duality, at least for compact S , see e.g. [14, 6.2.3.9 & 6.2.3.12]. However, one can also prove it directly by inductively choosing $M_n := \text{span}\{Sx_k : k < n\}$, $x_n \in B_X$ with $(1 + \varepsilon)\|Q_{M_n}^Y Sx_n\| \geq \|Q_{M_n}^Y S\| \geq d_n(S)$, and $b_n \in B_{Y'}$ with $\langle Sx_n, b_n \rangle = \|Q_{M_n}^Y Sx_n\|$ and $\langle Sx_k, b_n \rangle = 0$ for $k < n$. The remaining proof is as above. $\square$

Remark 5 The proof of Theorem 3 uses the determinant to relate the eigenvalues $\lambda_k(S_n)$ of S_n (which are the diagonal entries) with its singular numbers. Sometimes, the more general *Weyl's inequality* [21] from 1949 is used, which states that $\prod_{k=1}^n |\lambda_k(S)| \leq \prod_{k=1}^n a_k(S)$ for any compact $S \in \mathcal{L}(H, H)$, see also [14, 3.5.1]. This crucial step appears in all the proofs I am aware of that lead to the optimal factor n in the comparisons. Unfortunately, all these approaches use a whole collection of s-numbers, which does not allow for bounds for individual n .

Let us present an example from [5, 2.d.5] that shows that such product bounds between eigenvalues and s-numbers are to some extent best possible:

For $0 < \sigma < 1$, consider the matrix $T_n = (\delta_{j,i+1} + \sigma \cdot \delta_{i,n} \delta_{j,1})_{i,j=1}^n$ which represents a mapping on ℓ_2^n . It is easy to verify that $a_k(T_n) = 1$ for $k < n$ and $a_n(T_n) = \sigma$. (Recall that a_k are the singular numbers in this case.) Moreover, since $T_n^n = \sigma \cdot I_{\ell_2^n}$, we see that $|\lambda_k(T_n)| = \sigma^{1/n}$ for $k = 1, \dots, n$. This shows that Weyl's inequality, as well as the easy corollary $|\lambda_n(S)| \leq \|S\|^{1-\frac{1}{n}} a_n(S)^{1/n}$, are in general best possible.

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